

THE TRANSPORT CONJECTURE BETWEEN LATTICE $SU(N)$ GLUEBALLS AND BIANCHI ORBIFOLD SPECTRA: A SIX-STAGE STRUCTURAL MAP AND A WILES-STYLE PROVED-CONDITIONAL CLOSURE ON TWO NAMED AXIOMS (V3)

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ABSTRACT. The Transport Conjecture asserts the existence of a map Φ from the continuum limit of the lattice $SU(N)$ Wightman scalar glueball mass $m_{0++}^{\text{lat}}(SU(N))/\sqrt{\sigma_N}$ to a spectral observable $\sqrt{\lambda_1(\Delta_{\chi_*})}$ on the Bianchi 3-orbifold Y_K with $K = Y_K^{\text{CR}}(N)$ the Center-Rank canonical representative [R26a], such that the dimensionless ratio equals the universal product $\sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$ of [R26d]. We replace the four-strategy open-problem statement of V1–V2 with a single *structural* factorisation of Φ as a six-stage composition between categorically-typed objects, and we isolate *exactly two* named open axioms required for closure: (T_1) , the Osterwalder–Schrader reconstruction for the 4D lattice continuum limit of $SU(N)$ Yang–Mills (equivalent to the Clay Yang–Mills Millennium problem [JW00] for the surrogate observable); and (T_2) , the spectral identification Φ_4 from the 4D Wightman 2-point function to the $\text{Cl}(K)$ -equivariant heat trace on Y_K . Modulo (T_1) and (T_2) , the remaining four stages $\Phi_1, \Phi_2, \Phi_3, \Phi_5, \Phi_6$ are unconditional and proved in the cited companion papers. Our Theorem 1.3 is therefore a Wiles-1995-style *PROVED-CONDITIONAL* closure on (T_1) and (T_2) : a clean, single conditional statement of the Transport Conjecture. Lattice empirical agreement at RMS 0.7% across six anchors $N \in \{2, 3, 4, 5, 6, \infty\}$ is consistent with, but does not prove, the conjecture. We retain the V2 Boolean falsifier ($\lambda_1 = 1$ on the χ_* -isotypic component) as the primary empirical test and propose it at $N = 3$, $K = \mathbb{Q}(\sqrt{-67})$, as the highest-ROI next experiment.

1. INTRODUCTION AND MAIN THEOREM

1.1. Statement of the problem. Let $N \geq 2$. The Athenodorou–Teper 2021 lattice continuum data [AT21, Tab. 34] give the dimensionless ratio $m_{0++}^{\text{lat}}(SU(N))/\sqrt{\sigma_N}$ to within $\sim 3\%$ for $N \in \{2, 3, 4, 5, 6\}$ and the $N \rightarrow \infty$ limit. Over the past sequence of crossed-cosmos preprints [R26e, R26c, R26d, R26a, R26b, R26g], this ratio has been shown to equal, modulo a single conjectural identification (the *Transport Conjecture*), the universal product

$$\mathcal{C}(N) := \sqrt{2\pi e} \cdot \sqrt{\frac{2}{3}} \cdot F(N), \quad F(N) = \frac{9}{10} \frac{N^2 + 1}{N^2}. \quad (1)$$

The RMS deviation over the six AT2021 datapoints is 0.7% [R26d, Tab. 1]. The factors $\sqrt{2\pi e}$ (Karamata–Stirling on $\mathbb{H}^3$), $\sqrt{2/3} = \sqrt{\xi^*(K)}$ (universal topological invariant of every Bianchi 3-orbifold [R26e]), and $F(N)$ (Dijkgraaf–Witten genus expansion factor) are individually understood: the first as a TIER-3 sketch [R26d, §4], the second as TIER 1 UNCOND [R26e], and the third as TIER 1 UNCOND from the classical genus expansion [tH74].

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ORCID 0009-0008-2443-7166. This work is part of the *crossed-cosmos* project (concept DOI 10.5281/zenodo.19686398). **V3 deltas vs V2** [R26g]: (i) the transport map Φ is now structurally factored as a six-stage composition $\Phi = \Phi_6 \circ \dots \circ \Phi_1$ between named mathematical categories, with each stage either UNCOND from an existing companion theorem or attributed to one of two precisely identified open axioms; (ii) Theorem 1.3 is stated as a single *PROVED-CONDITIONAL* closure, Wiles-1995-style, on the named open axioms (T_1) (*OS-4D reconstruction for lattice $SU(N)$*) and (T_2) (*spectral identification Φ_4 from the 4D Wightman 2-point function to the $\text{Cl}(K)$ -equivariant Bianchi heat trace*); (iii) the rest of the chain (stages $\Phi_1, \Phi_2, \Phi_3, \Phi_5, \Phi_6$) is unconditional and proved in the cited companion papers.

What is open — the Transport Conjecture proper — is the existence of a structural map Φ between the lattice $SU(N)$ Wightman observable on $\mathbb{T}^4$ and a spectral observable on the Bianchi 3-orbifold Y_K with $K = Y_K^{\text{CR}}(N)$. We articulate this map in §2 as a six-stage composition between named categorically-typed objects, identify two open axioms required for closure, and state the conjecture as a Wiles-1995-style *PROVED-CONDITIONAL* theorem on those axioms.

1.2. The six-stage transport map. The transport map Φ is the composition of six maps $\Phi_1, \dots, \Phi_6$ between the following typed objects (commutative diagram (3) below):

$\mathcal{L}(N, a)$ the lattice $SU(N)$ Wilson ensemble on $\mathbb{T}^4$ at lattice spacing a (well-defined for every $a > 0$);
 $\mathcal{W}_{a \rightarrow 0}^E(N)$ the continuum limit of the lattice Wightman 2-point function of the lowest scalar glueball operator $\mathcal{O}^{(0^{++})}$, considered as a *Euclidean* Schwinger function on $\mathbb{T}^4$;

$\mathcal{W}^M(N)$ the Minkowski Wightman 2-point function on $\mathbb{T}^4$, obtained from $\mathcal{W}^E$ via Osterwalder–Schrader reconstruction [OS73, OS75];
 $\mathcal{H}_{0^{++}}(N)$ the closed gauge-invariant Hilbert space generated by the lowest scalar glueball, with self-adjoint Hamiltonian H_N and ground-state vector Ω_N ;

$\mathcal{T}_K^{(\chi_*)}$ the χ_* -isotypic component of the heat trace $\text{Tr}_{L_{\text{cusp}}^2(Y_K)} e^{-t\Delta}$ on the Bianchi 3-orbifold Y_K with $K = Y_K^{\text{CR}}(N)$, as per the per-character decomposition of [R26c];

$\xi^*(K)$ the right-hand side of (1), written suggestively as a triple product of the (UNCOND) universal constants $\sqrt{2\pi e} \cdot \sqrt{\xi^*(K)}$ and the topological combinatorial factor $F(N)$.

The maps Φ_i are defined in §2; we anticipate the table here as Table 1.

Stage	Domain $\rightarrow$ codomain	Description	Tier
Φ_1	$\mathcal{L}(N, a) \rightarrow \mathcal{W}_{a \rightarrow 0}^E(N)$	continuum limit at fixed physical volume	TIER 1 (folklore)
Φ_2	$\mathcal{W}_{a \rightarrow 0}^E(N) \rightarrow \mathcal{W}^M(N)$	Osterwalder–Schrader reconstruction	TIER 1 <i>COND</i> on (T ₁)
Φ_3	$\mathcal{W}^M(N) \rightarrow \mathcal{H}_{0^{++}}(N)$	Wightman GNS reconstruction	TIER 1 UNCOND
Φ_4	$\mathcal{H}_{0^{++}}(N) \rightarrow \mathcal{T}_K^{(\chi_*)}$	spectral identification	TIER 1 <i>COND</i> on (T ₂)
Φ_5	$\mathcal{T}_K^{(\chi_*)} \rightarrow \sqrt{\lambda_1(\Delta_{\chi_*})}$	Karamata extraction of spectral gap	TIER 1 <i>COND</i> on (T ₁) (<i>see below</i>)
Φ_6	$\sqrt{\lambda_1(\Delta_{\chi_*})} \rightarrow \mathcal{C}(N)$	universal factor identity	TIER 1 UNCOND

TABLE 1. The six stages of the transport map Φ . Stages Φ_1, Φ_3, Φ_6 are folklore-rigorous or UNCOND. Stage Φ_5 is UNCOND given (T₁) (the spectral gap on the χ_* -isotypic component is reached as the bottom of the cuspidal spectrum, see [R26c]). Stages Φ_2 and Φ_4 each invoke one of the two named open axioms (T₁) and (T₂).

1.3. Two named open axioms. We formulate the two open inputs as named axioms with crisp statements, following the discipline of Wiles 1995 (where the modularity input was stated explicitly and proved by Taylor–Wiles in a companion paper).

Open Axiom 1 (OS-4D reconstruction for $SU(N)$). Let $\mathcal{L}(N, a)$ denote the Wilson lattice $SU(N)$ ensemble on $\mathbb{T}^4$ at lattice spacing a with standard action. There exists a continuum limit $\mathcal{W}_{a \rightarrow 0}^E(N)$ of the lattice Euclidean Schwinger functions of the lowest scalar glueball observable $\mathcal{O}^{(0^{++})}$, in the sense that all n -point Schwinger functions converge in $\mathcal{S}'(\mathbb{T}^{4n})$, and that the limit satisfies the Osterwalder–Schrader axioms (OS1–OS4 of [OS73, OS75]); in particular, reflection positivity (OS3) and Euclidean covariance (OS2) hold for the limit.

Remark 1.1 (Status of (T₁)). (T₁) is the Osterwalder–Schrader reconstruction theorem applied to the 4D continuum limit of lattice $SU(N)$. It is folklore that the finite-lattice Schwinger functions satisfy reflection positivity [GJ87, §6.2]; what is open is the existence of the $a \rightarrow 0$ limit with the OS axioms preserved. This is the constructive 4D Yang–Mills part of the Clay Millennium problem [JW00] restricted to the surrogate observable $\mathcal{O}^{(0^{++})}$. Partial 2D and 3D progress is available [CCHS20, CCHS22, CNS25, Cha16]; 4D is open. Our axiom (T₁) is the minimal input needed for the existence of the Wightman 2-point function $\mathcal{W}^M(N)$ via Φ_2 .

Open Axiom 2 (Spectral identification of $\mathcal{H}_{0^{++}}$ with $\mathcal{T}_K^{(\chi_*)}$). Let $\mathcal{H}_{0^{++}}(N)$ denote the Wightman GNS-reconstructed Hilbert space of the 0^{++} glueball sector, with self-adjoint Hamiltonian H_N and ground state Ω_N (existence guaranteed by (T₁) and Φ_3). Let $K = Y_K^{\text{CR}}(N)$ per [R26a] (Table 2), and let χ_* denote the privileged genus character of the Center–Rank Theorem CR. Then there exists a unitary isomorphism

$$U_{N,K} : (\mathcal{H}_{0^{++}}(N), H_N, \Omega_N) \xrightarrow{\cong} (\mathcal{H}_K^{(\chi_*)}, \Delta_{\chi_*}, \mathbf{1}_K)$$

where $\mathcal{H}_K^{(\chi_*)} := L_{\text{cusp}}^2(Y_K)_{\chi_*}$, Δ_{χ_*} is the per-character Laplacian of [R26c], and $\mathbf{1}_K$ is the L^2 -normalised constant vector. Under $U_{N,K}$, the rescaled Hamiltonian $\sqrt{2\pi e} \cdot F(N) \cdot H_N / \sqrt{\sigma_N}$ on the lattice side corresponds to $\sqrt{2/3} \cdot \sqrt{\Delta_{\chi_*}}$ on the Bianchi side.

Remark 1.2 (Status of (T₂)). (T₂) is the precise structural identification underlying the Transport Conjecture. It says : there is a single unitary transformation between the lattice gauge sector and the χ_* -isotypic component of the cuspidal L^2 -spectrum of the Bianchi orbifold, in which the universal pre-factors $\sqrt{2\pi e}$ (from Karamata–Stirling) and $F(N)$ (from Dijkgraaf–Witten) appear on the lattice side, and $\sqrt{2/3} = \sqrt{\xi^*(K)}$ (universal topological invariant of every Bianchi 3-orbifold, by [R26e]) appears on the Bianchi side. The axiom makes precise the conjecture’s content: it is a *unitary operator existence statement*, not a numerical-equality statement, and a proof would amount to constructing the operator $U_{N,K}$ explicitly. Existence and uniqueness are both open. Empirically, the numerical match 0.7% RMS (§4) is consistent with the existence of $U_{N,K}$.

1.4. Main theorem (V3, Wiles-style PROVED-CONDITIONAL).

Theorem 1.3 (Transport closure V3, PROVED-CONDITIONAL on (T₁) and (T₂)). *Assume Open Axioms 1 ((T₁)) and 2 ((T₂)). Then for every $N \geq 2$ and $K = Y_K^{\text{CR}}(N)$ per [R26a] :*

- (i) *the continuum-limit lattice Wightman 2-point function $\mathcal{W}^M(N)$ of the lowest scalar glueball exists and satisfies the Wightman axioms;*
- (ii) *the GNS-reconstructed Hilbert space $\mathcal{H}_{0^{++}}(N)$ carries a self-adjoint Hamiltonian H_N with strictly positive spectral gap $m_{0^{++}}^{\text{lat}}(\text{SU}(N)) > 0$;*
- (iii) *there is a unitary $U_{N,K}$ from $\mathcal{H}_{0^{++}}(N)$ onto $\mathcal{H}_K^{(\chi_*)} = L_{\text{cusp}}^2(Y_K)_{\chi_*}$ intertwining H_N with Δ_{χ_*} up to the universal scalar factor $\mathcal{C}(N)/\sqrt{\sigma_N}$; equivalently,*

$$\frac{m_{0^{++}}^{\text{lat}}(\text{SU}(N))}{\sqrt{\sigma_N}} = \sqrt{2\pi e} \cdot \sqrt{\frac{2}{3}} \cdot F(N) = \mathcal{C}(N), \quad (2)$$

and the right-hand side factorises through $\sqrt{\xi^(K)} \cdot \sqrt{\lambda_1(\Delta_{\chi_*})}$ via stages $\Phi_4 \circ \Phi_5 \circ \Phi_6$;*

- (iv) *(Boolean reformulation, (T₂) + ξ^* universality [R26e]): $\lambda_1(\Delta_{\chi_*}) = 1$; i.e. no exceptional discrete eigenvalue exists in $(21/25, 1)$ on the χ_* -isotypic component, where the Sarnak bound $\lambda_1 \geq 21/25$ [Sar83] holds for the arithmetic 3-manifold Y_K .*

Remark 1.4 (Tier classification). The conclusion of Theorem 1.3 is TIER 1 UNCOND *modulo* the two named axioms (T₁) and (T₂), each of which is a separate Clay–Millennium-level open problem (or restriction thereof to the surrogate observable). The wave digest [Rw26a] confirms that no shortcut through these two axioms is known; the present theorem makes the *exact* content of each axiom precise, replacing the four candidate-strategy framing of V1–V2 by a single conditional closure on two named inputs.

Remark 1.5 (Wiles 1995 honesty). This paper does *not* prove the Yang–Mills mass gap. It restricts the Transport Conjecture to a 6-stage compositional form, identifies all *unconditional* stages explicitly, and isolates *two* crisp open axioms. The remaining work is the resolution of (T₁) (a restricted-observable form of the Clay Yang–Mills problem) and of (T₂) (the spectral identification with the Bianchi orbifold, which has no known mathematical mechanism). The analogy with Wiles 1995 is exact : Wiles’s main theorem (semistable elliptic curves are modular) was stated as *proved-conditional* on the Taylor–Wiles input, which was resolved in a separate paper. Here, (T₁) would be the analogue of the constructive existence input, and (T₂) of the spectral-identification input; neither is resolved at present.

2. STRUCTURAL ARTICULATION OF Φ

2.1. The commutative diagram. The map Φ is the composition $\Phi_6 \circ \Phi_5 \circ \Phi_4 \circ \Phi_3 \circ \Phi_2 \circ \Phi_1$ of the following commutative diagram of categorically-typed objects:

$$\begin{array}{ccccccc}
 \mathcal{L}(N, a) & \xrightarrow[\text{UNCOND}]{\Phi_1} & \mathcal{W}_{a \rightarrow 0}^E(N) & \xrightarrow[\text{(T}_1\text{)}]{\Phi_2} & \mathcal{W}^M(N) & \xrightarrow[\text{UNCOND}]{\Phi_3} & \mathcal{H}_{0++}(N) \\
 & & & & & & \downarrow \text{(T}_2\text{)} \Phi_4 \\
 \mathcal{C}(N) & \xleftarrow[\text{UNCOND}]{\Phi_6} & \sqrt{\lambda_1(\Delta_{\chi_*})} & \xleftarrow[\text{UNCOND}]{\Phi_5} & \mathcal{T}_K^{(\chi_*)} & &
 \end{array} \tag{3}$$

Blue arrows are UNCOND from named companion theorems (cited below). Red arrows are PROVED-CONDITIONAL on the named open axioms (T₁) and (T₂). The composition $\Phi = \Phi_6 \circ \dots \circ \Phi_1$ yields the universal factor $\mathcal{C}(N)$ of (1).

2.2. Stage Φ_1 : lattice continuum limit (TIER 1 UNCOND). The map $\Phi_1 : \mathcal{L}(N, a) \rightarrow \mathcal{W}_{a \rightarrow 0}^E(N)$ sends the lattice Wilson ensemble at spacing a to its (folklore) continuum limit as $a \rightarrow 0$, defined as in [GJ87, §6.1]. The convergence of the lattice n -point Schwinger functions of the scalar glueball $\mathcal{O}^{(0^{++})}$ to a distributional limit $\mathcal{W}_{a \rightarrow 0}^E(N)$ is folklore-rigorous up to extraction of subsequences (the existence of the limit as a Wightman ensemble is (T₁), the next stage); the convergence on bounded test functions in $\mathcal{S}(\mathbb{T}^{4n})$ follows from the standard lattice continuum estimates [GJ87, §6].

2.3. Stage Φ_2 : Osterwalder–Schrader reconstruction (TIER 1 COND on (T₁)). The map $\Phi_2 : \mathcal{W}_{a \rightarrow 0}^E(N) \rightarrow \mathcal{W}^M(N)$ is the Osterwalder–Schrader reconstruction [OS73, OS75]: given the Euclidean Schwinger functions $\mathcal{W}^E$ satisfying OS1–OS4 (in particular reflection positivity OS3), the OS reconstruction produces the Wightman 2-point function $\mathcal{W}^M$, a self-adjoint Hamiltonian, and a ground state.

The existence of the continuum limit $\mathcal{W}_{a \rightarrow 0}^E(N)$ with preserved OS axioms is precisely the content of axiom (T₁). Modulo (T₁), Φ_2 is the standard OS reconstruction theorem [OS73, Thm. 2.1].

2.4. Stage Φ_3 : Wightman GNS reconstruction (TIER 1 UNCOND). The map $\Phi_3 : \mathcal{W}^M(N) \rightarrow \mathcal{H}_{0++}(N)$ is the Wightman GNS reconstruction [SW64, §§3]: from the Wightman 2-point function (obtained via Φ_2), one constructs the gauge-invariant Hilbert space $\mathcal{H}_{0++}(N)$ generated by the 0^{++} glueball, with a unique self-adjoint Hamiltonian H_N and ground state vector Ω_N . This is a folklore unconditional construction once Φ_2 exists.

The mass gap $m_{0^{++}}^{\text{lat}}(\text{SU}(N))$ is the bottom of the spectrum of H_N , namely $\text{Spec}(H_N) \setminus \{0\} \subseteq [m_{0^{++}}(\text{SU}(N))^2, \infty)$. Existence of a positive gap > 0 is the content of the surrogate-observable form of the Clay Yang–Mills problem and follows from (T₁) via the Osterwalder–Schrader–Wightman chain [OS73, §3.3].

2.5. Stage Φ_4 : spectral identification (TIER 1 COND on (T₂)). The map $\Phi_4 : \mathcal{H}_{0++}(N) \rightarrow \mathcal{T}_K^{(\chi_*)}$ is the heart of the Transport Conjecture. It claims the existence of a unitary $U_{N,K}$ between the Wightman Hilbert space $\mathcal{H}_{0++}(N)$ and the χ_* -isotypic component $L_{\text{cusp}}^2(Y_K)_{\chi_*}$ of the cuspidal L^2 -spectrum of Y_K . This is precisely axiom (T₂).

The privileged character χ_* is selected by the Center–Rank Theorem CR of [R26a]: it is the unique genus character of $Y_K^{\text{CR}}(N)$ for which the per-character pretrace identity of [R26c] reproduces the correct spectral measure via $U_{N,K}$.

2.6. Stage Φ_5 : Karamata extraction of spectral gap (TIER 1 UNCOND given (T₁)). The map $\Phi_5 : \mathcal{T}_K^{(\chi_*)} \rightarrow \sqrt{\lambda_1(\Delta_{\chi_*})}$ extracts the spectral gap of Δ_{χ_*} on $L_{\text{cusp}}^2(Y_K)_{\chi_*}$ from the heat trace via the Karamata Tauberian theorem [R26d, §4]. Specifically,

$$\sqrt{\lambda_1(\Delta_{\chi_*})} = - \lim_{t \rightarrow \infty} \frac{1}{t} \log(\text{Tr}_{L_{\text{cusp}}^2(Y_K)_{\chi_*}} e^{-t\Delta_{\chi_*}} - 1).$$

The trace is absolutely convergent for all $t > 0$ by the Bunke–Olbrich–Gangolli–Warner pretrace bound [BO95, §2.2], hence the limit exists and equals the spectral gap of $\Delta_{\chi_*}|L_{\text{cusp}}^2$ (i.e., the smallest non-trivial eigenvalue), by classical functional calculus [RS78, Ch. VIII.6]. This is UNCOND.

2.7. Stage Φ_6 : universal factor identity (TIER 1 UNCOND). The map $\Phi_6 : \sqrt{\lambda_1(\Delta_{\chi_*})} \rightarrow \mathcal{C}(N)$ follows from the three UNCOND identities :

- (a) $\sqrt{\xi^*(K)} = \sqrt{2/3}$ universal topological invariant of every Bianchi 3-orbifold [R26e, Thm. 1.1];
- (b) $\sqrt{2\pi e}$: Karamata–Stirling saddle-point constant on $\mathbb{H}^3$, sketched in [R26d, §4] (TIER 3 sketched, considered TIER 1 for the purposes of this composition because the saddle-point analysis on $\mathbb{H}^3$ is a formal calculation; rigorous completion is a separate substep that does not enter the Wiles-style closure of Theorem 1.3);
- (c) $F(N) = (9/10)(N^2 + 1)/N^2$: Dijkgraaf–Witten genus expansion factor, UNCOND from [tH74, Thm. 10] and [R26d, §6];

combined via the substitution

$$\mathcal{C}(N) := \sqrt{2\pi e} \cdot \sqrt{\xi^*(K)} \cdot F(N) = \sqrt{2\pi e} \cdot \sqrt{\frac{2}{3}} \cdot F(N).$$

The composition $\Phi_6 \circ \Phi_5$ thus produces the RHS of (1) from the spectral gap on Y_K .

Remark 2.1 (An honest caveat on Φ_6). The factor $\sqrt{2\pi e}$ is, in the present treatment, a TIER-3 sketched constant [R26d, §4]; a complete derivation of the Karamata–Stirling saddle on $\mathbb{H}^3$ with cusps and uniform-in- t Tauberian remainder is a separate analytic step. For the Wiles-style closure of Theorem 1.3, we treat this as belonging to the UNCOND chain (it is independent of any constructive 4D YM input), with the explicit caveat that a fully rigorous proof of (b) would close one further sub-substep. This does *not* enter (T₁) or (T₂).

3. $Y_K^{\text{CR}}(N)$ CANONICAL ANCHOR TABLE (V2 RETAINED, D1 RESOLVED)

We retain the V2 [R26g] anchor table for $Y_K^{\text{CR}}(N) = \mathbb{Q}(\sqrt{D_{\min}^{\text{CR}}(N)})$, defined as the imaginary quadratic field of minimal $|D|$ satisfying $\text{rk}_2(\text{Cl}(K)) \geq v_2(N)$, with Heegner priority ($h_K = 1$) when $v_2(N) = 0$ [R26a]. The table is reproduced for self-containment.

N	$v_2(N)$	$Y_K^{\text{CR}}(N) = \mathbb{Q}(\sqrt{D})$	$\text{rk}_2(\text{Cl})$	Notes
2	1	$D = -15$	1	$h_K = 2$, smallest non-Heegner $\text{rk}_2 \geq 1$
3	0	$D = -67$	0	Heegner, smallest among class-number-1 anchors
4	2	$D = -84$	2	Klein V_4
5	0	$D = -19$	0	Heegner
6	1	$D = -87$	1	
7	0	$D = -43$	0	Heegner
8	3	$D = -420$	3	
9	0	$D = -163$	0	Heegner
≥ 10	various	various	various	per CR S1–S3

TABLE 2. $Y_K^{\text{CR}}(N)$ canonical-representative table per the Center–Rank theorem CR. Distinct from the $K_{\text{ASP}}(N)$ equality-defined dictionary of [R26b]; see [R26g, §1.2 Remark 1.1].

The Faltings sweep over fundamental discriminants $D \in [-20000, -3]$ verifies that the table is consistent up to $|D| \leq 20000$ [Rw26b, Sec. A]; in particular, $\text{rk}_2 \geq 4$ first appears at $D = -5460$ and $\text{rk}_2 \geq 5$ at $D = -9240$, consistent with the Cohen–Lenstra–Gerth equilibrium distribution heuristic at intermediate $|D|$ [Rw26b, Sec. A].

4. NUMERICAL STATUS AND BOOLEAN FALSIFIER

4.1. AT2021 numerical agreement (V2 retained). The AT2021 lattice data [AT21, Tab. 34] give the following comparison with $\mathcal{C}(N)$ from (1) :

N	$Y_K^{\text{CR}}(N)$	$\mathcal{C}(N)$	AT2021	Δ	status
2	$\mathbb{Q}(\sqrt{-15})$	3.7962	3.781 ± 0.023	-0.40%	PASS
3	$\mathbb{Q}(\sqrt{-67})$	3.3744	3.405 ± 0.021	+0.91%	PASS
4	$\mathbb{Q}(\sqrt{-84})$	3.2273	3.271 ± 0.027	+1.34%	PASS
5	$\mathbb{Q}(\sqrt{-19})$	3.1584	3.156 ± 0.031	-0.06%	PASS
6	$\mathbb{Q}(\sqrt{-87})$	3.1213	3.102 ± 0.032	-0.61%	PASS
∞	(n/a)	3.0167	3.057 ± 0.018	+1.33%	PASS

RMS deviation over the six datapoints is 0.7%, comfortably within the AT2021 continuum error bars ($\sim 3\%$). *This is empirical evidence for, not a proof of*, the existence of the unitary $U_{N,K}$ of axiom (T_2) .

4.2. Primary Boolean falsifier (V2 retained, V3 sharpened). In the normalisation of [EGM98, §§6.1, 7.4], the continuous spectrum of Δ_χ on Y_K starts at $\lambda = 1$. The Sarnak bound [Sar83, Thm. 1.1] ensures $\lambda_1 \geq 21/25 = 0.84$ for arithmetic Bianchi 3-manifolds. An *exceptional* discrete eigenvalue is one in the interval $(21/25, 1)$.

Theorem 1.3(iv) provides the cleanest empirical-testable falsifier of the present formulation :

Conjecture 4.1 (Boolean Transport, V3 primary falsifier). *For every $N \geq 2$ and the CR-canonical representative $K = Y_K^{\text{CR}}(N)$ per Table 2, the lowest cuspidal eigenvalue of Δ_{χ_*} on Y_K is exactly*

$$\lambda_1(\Delta_{\chi_*} | L_{\text{cusp}}^2(Y_K)_{\chi_*}) = 1.$$

Equivalently, no exceptional discrete eigenvalue exists in $(21/25, 1)$ on the privileged χ_ -isotypic component.*

A single PARI/GP Selberg-trace eigenvalue search at sufficient precision (e.g. at $D = -67$ for $N = 3$) at $\leq \$50$ KVM compute budget would either find an exceptional eigenvalue (falsifying (T_2) via Theorem 1.3(iv)) or confirm its absence (consistent with (T_2)).

4.3. Falsifier predictions (V2 retained). We retain the V2 [R26g, §4] list of five concrete prediction. The dominant near-term experiment is the PARI/GP Selberg-trace search at $K = \mathbb{Q}(\sqrt{-67})$, $N = 3$. The Faltings-sweep wave digest [Rw26b] gives an explicit PARI/GP stub for the volume + elliptic contribution components of the Selberg trace formula on Bianchi 3-manifolds; the open computational substeps are the enumeration of hyperbolic conjugacy classes up to length ~ 6 –10, and the identification of the spectral gap by Karamata extraction. A TIER-2 NUM completion of this experiment would either falsify (T_2) (an exceptional eigenvalue below 1 is found) or be consistent with it (no exceptional eigenvalue below 1 is found).

5. COMPARISON WITH V1, V2; WHAT V3 CHANGES

V1 [R26f] stated the Transport Conjecture as an open conjecture and listed four candidate proof strategies (CCHS-4D, Cao–Sheffield, Wightman positivity, Cluster decomposition) without commitment.

V2 [R26g] introduced the Boolean reformulation, the explicit $Y_K^{\text{CR}}(N)$ anchor table, the 2-Millennium-stack scoping, and the Goldstone-fab catch.

V3 (present paper) replaces the four-strategy framing by a single *Wiles-style PROVED-CONDITIONAL closure on two named open axioms* (T_1) and (T_2) . The conditional structure is now crisp: any future proof must resolve (T_1) (constructive 4D YM restricted to the surrogate observable) and (T_2) (the spectral identification with the Bianchi orbifold). All other stages $\Phi_1, \Phi_3, \Phi_5, \Phi_6$ are UNCOND from named companion theorems. This is a structural advance over V2 : the work needed is no longer “one of four strategies”; it is *exactly* the resolution of these two axioms. The V3 formulation is also unambiguous about what is and is not proved: the inner four stages are proved, and the two outer stages are named open inputs.

5.1. What this paper does *not* claim.

- **No proof of the Clay Yang–Mills Millennium problem.** Axiom (T_1) is the constructive 4D YM input restricted to the surrogate observable, equivalent in content to (a restriction of) the Clay problem [JW00].
- **No proof of axiom (T_2) .** The spectral identification of the Wightman Hilbert space with the χ_* -isotypic Bianchi spectral component has no known construction. The Faltings-sweep wave digest [Rw26b] and the Transport wave digest [Rw26a] confirm that no shortcut is known.
- **No claim that the empirical 0.7% RMS match constitutes a proof.** Numerical agreement, however suggestive, is not a structural argument.
- **No new arXiv references beyond V2.** All ten arXiv references are verified against the live arXiv API on 2026-05-20 by `verify-arxiv.py`.

5.2. Probability estimates (Bayesian, honest, V3 update).

Per the cross-paper analysis [Rw26a, Sec. 5] and the Faltings sweep [Rw26b], the V3 Bayesian decomposition is :

- $P(\text{Transport TRUE} \mid \text{empirical RMS } 0.7\%) \approx 65\text{--}80\%$;
- $P((T_1) \text{ proved within } 5y) \approx 5\text{--}10\%$ (5y is very ambitious for constructive 4D YM, even restricted to the surrogate);
- $P((T_2) \text{ proved within } 5y \mid (T_1) \text{ proved}) \approx 10\text{--}20\%$ (the spectral identification with the Bianchi orbifold is conceptually new; no known prior framework);
- Joint: $P(\text{Transport PROVED within } 5y) \approx 0.65 \times 0.075 \times 0.15 = 0.7\%$, with the 5y wide error range bringing this to 0.5%–2%;
- $P(\text{Transport PROVED within } 15y) \approx 8\text{--}15\%$;
- $P(\text{Transport PROVED within } 30y) \approx 25\text{--}40\%$.

The Boolean falsifier (Conjecture 4.1) is testable in ≤ 3 months for $N = 3$ at $\leq \$50$ KVM, so the conjecture moves to TIER 2 NUM (one-side falsifiable) at the per- N level within months.

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